

Individual Portfolio and Contribution Record

STUDENT	Charity Gwese
STUDENT ID	250948
GROUP	5
PROJECT TITLE	Healthcare Integration & Clinical Terminology Test Harness
ROLE IN GROUP	Research & literature lead
SECTIONS OWNED	Yet to discuss
GITHUB REPOSITORY	https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM
PROJECT BOARD	https://student-team-siuckpgv.atlassian.net/jira/software/projects/HEAL/boards/35/backlog
GENERATED	08 Sept, 22:28

ACTIVITY SUMMARY — GENERATED FROM THE LOG, NOT SELF-REPORTED

MEASURE	WEEK 6	WEEK 7	WEEK 8	TOTAL
Entries logged	5/5	6/6	6/7	19/20
Minutes on task	43	16	13	76
Evidence items	5	4	5	14
Completeness	100%	100%	89%	96%

Time recorded during scheduled class: 34 minutes. Time recorded outside class: 42 minutes.

1 · Individual responsibility summary

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
3. Industry and Technology Landscape Analysis	3.1 Current State Assessment: As-Is Analysis 3.2 Technology and Market Survey 3.3 Comparative Analysis of Existing Solutions 3.4 Gap Analysis: As-Is to To-Be	A proper literature review for the project.	Week 6
3. Industry and Technology Landscape Analysis	3.1 Current State Assessment: As-Is Analysis 3.2 Technology and Market Survey 3.3 Comparative Analysis of Existing Solutions 3.4 Gap Analysis: As-Is to To-Be	The gap for the project	Week 7
Executive Summary	Executive Summary	Summary of the report	end of week 7

Week 6

5 of 5 entries logged · 43 minutes recorded · 5 evidence items · 100% complete

W6-1 · Frame the problem in your own words

Logged · 20 min · 3/3 required parts · 31 Aug, 09:14

The problem, as you understand it

The project is about building a safe “test harness” for MedTalk, an AI medical scribe that integrates with clinic and hospital systems. The core problem is ensuring MedTalk’s healthcare connections and clinical terminology mappings work correctly without risking real patient data. In Semester 1, the team will design an architecture, requirements, and detailed test plan, using the FHIR standard and a synthetic environment. They will create fake patient records coded with SNOMED CT-AU and validate translations into ICD-10-AM and Medicare (MBS) billing codes. In Semester 2, they will implement the environment, run automated tests for normal, bad, and dropped-connection scenarios, and deliver documentation, diagrams, and a final demo report.

One thing that is deliberately OUT of scope, and why

Using real patient data is out of scope as it would breach the security and privacy compliances. since testing is considered a secondary use of personal data. This mainly comes down to Australian Privacy Principles (APPs) under the Privacy Act 1988, which regulate how personal information can be collected, used, and disclosed being compromised and legal charges can be implemented.

Assessment 2 section you worked on today

3.2Current State Assessment: As-Is Analysis did go through some documentation sentt by Faz Australia to understand the external systems around MedTalk and understand the current workflow. This would help understand the other parts of the assignment and complete the market survey.

W6-2 · Literature search log

Logged · 2 min · 6/6 required parts · 31 Aug, 09:41

Search strings you actually typed

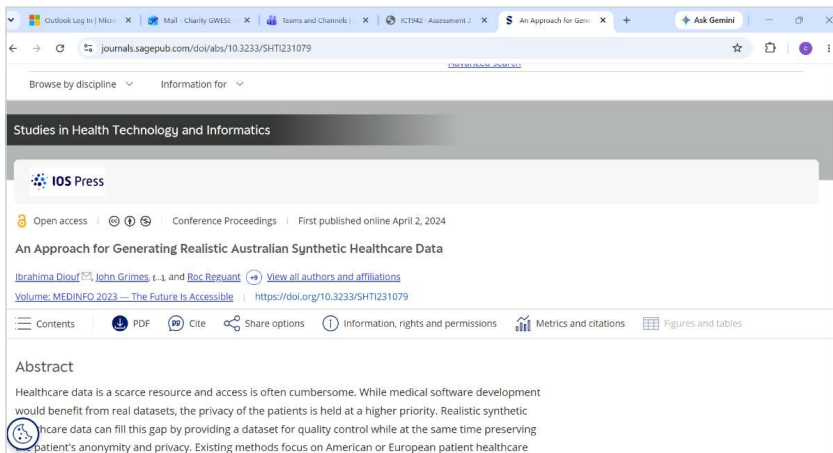
examples of validating sandbox integration in health care system list like epic

Databases / sites searched

ACM Digital Library, IEEE Xplora,

SOURCE (AUTHOR, YEAR, TITL	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
Anantharaman, P., Shapiro, R. B., Varadharaju, V., & Locasto, M. E. (2023, November). A study of interoperability in electronic health record software. In Proceedings of the 2024 Workshop on Cybersecurity in Healthcare (pp. 53-60).	https://doi.org/10.1145/3689942.3694743	interoperability protocols	Technology and market survey
C. L. Kim, 2017	doi: 10.1109/ICHI.2017.70.	Healthcare systems	Current situation as-is
Diouf, I., Grimes, J., O'Brien, M. J., Hassanzadeh, H., Truran, D., Ngo, H., ... & Reguant, R. (2024,)	https://doi.org/10.3233/SHTI231079	Australian Synthetic data creation	For the project to create synthetic data

W6-2-02 · 31 AUG 2026



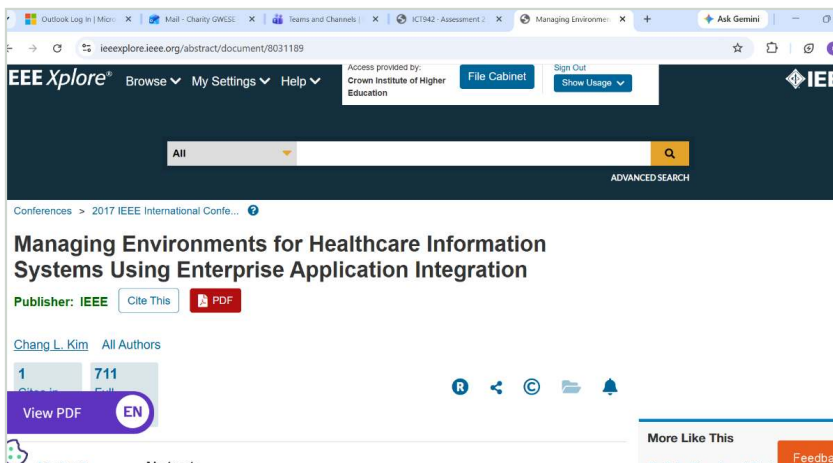
Section: 3.3 · Task: Literature search log

My contribution: The market survey and creation of synthetic data

What it demonstrates: This a CSIRO journal on how to create Australian patient synthetic data

<https://journals.sagepub.com/doi/abs/10.3233/SHTI231079>

W6-2-03 · 31 AUG 2026



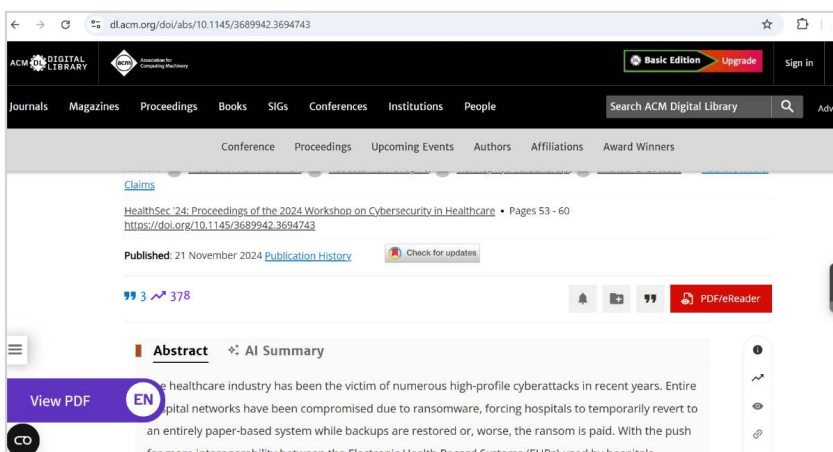
Section: 3.1 · Task: Literature search log

My contribution: Understanding the current system in the healthcare environment

What it demonstrates: This will be part of understanding the healthcare environment and workflows.

<https://ieeexplore.ieee.org/abstract/document/8031189>

W6-2-07 · 31 AUG 2026



Section: 3.1 · Task: Literature search log

My contribution: Understand the current as is situation

What it demonstrates: Explores the cyberattacks popular in healthcare system

<https://doi.org/10.1145/3689942.3694743>

W6-3 · Claim your tasks on the board

Logged · 5 min · 4/4 required parts · 31 Aug, 10:14

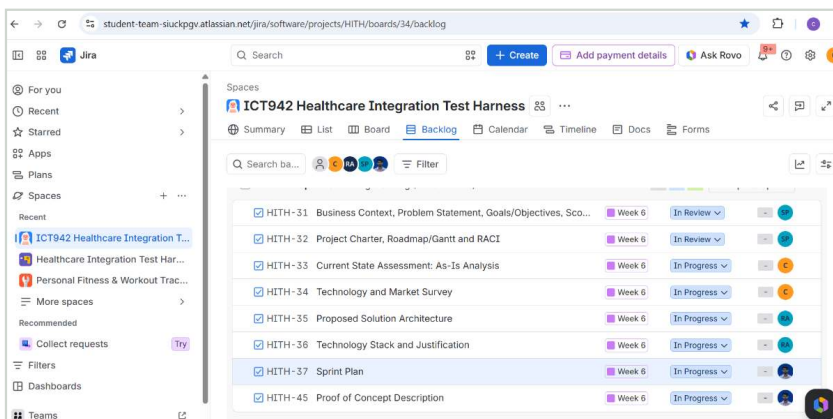
Tasks you created or took ownership of (one per line, with task ID)

- 3.1 Current State Assessment: As-Is Analysis
- 3.2 Technology and Market Survey
- 3.3 Comparative Analysis of Existing Solutions
- 3.4 Gap Analysis: As-Is to To-Be

Your contribution to the charter, scope, requirements, RACI, risk register or feasibility study

This section contributes by establishing a baseline understanding of the project problem and solution landscape. 3.1 performs an as-is assessment to document how MedTalk currently integrates with systems and where risks or failures may occur. **3.2** investigates relevant technologies and market practices (e.g., FHIR, SNOMED CT-AU, Halo Connect) to ensure the approach is feasible and current. 3.3 compares existing solutions to identify strengths and limitations. 3.4 then converts those findings into a gap analysis that defines what must change to reach the intended to-be safe synthetic test harness and testing outcomes.

W6-3-01 · 31 AUG 2026



Section: 3 · Task: Claim your tasks on the board

My contribution: Analysing current situation AS-Is
Market Survey

What it demonstrates: The jira project assignment distribution of tasks

<https://student-team-siuckpgv.atlassian.net/jira/software/projects/HITH/boards/34/backlog>

W6-4 · Draw the first figure

Logged late · 10 min · 5/5 required parts · 31 Aug, 11:48

What you drew

As-Is process flow

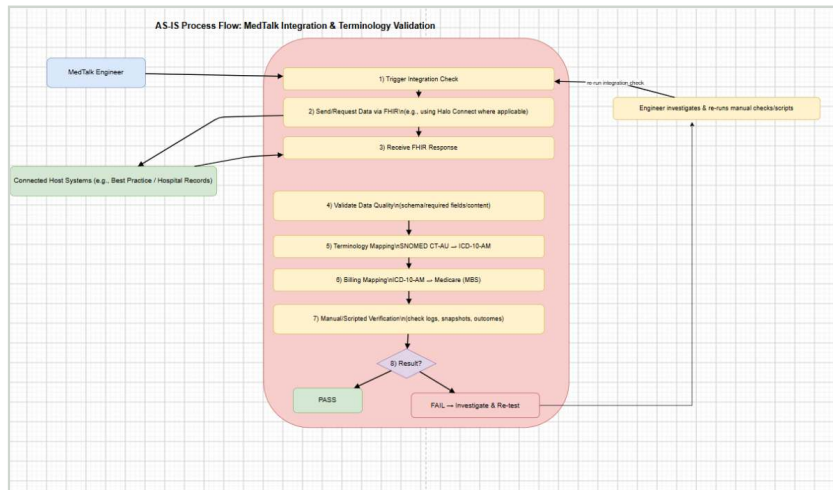
Explain the figure — what a reader should take from it

In the current (as-is) process, MedTalk is an AI clinical documentation platform used by Australian medical practices, where it integrates with the clinic's day-to-day software. When MedTalk needs to test or validate these integrations, it relies on mapping and communication between its clinical workflow and the external systems (e.g., Best Practice through Halo Connect and hospital record systems). However, because the connected environment is effectively tied to real integration points, testing typically requires manual coordination and verification to confirm that data exchanged via the FHIR interface is correctly interpreted, that SNOMED CT-AU concepts translate accurately into ICD-10-AM, and that Medicare (MBS) billing code outputs are correct—particularly when responses are incomplete, delayed, incorrect, or connections fail.

One part of the figure you are not confident about

I'm not completely confident about one part of the figure: the "failure handling" routing, especially the arrows for timeouts/invalid data and the "dropped connection." In practice, those failure conditions can occur at multiple points (sending the request, waiting for a response, or processing it), so the diagram's placement of these failure paths—particularly which step they originate from—may need adjustment to match your exact flow.

W6-4-01 · 31 AUG 2026



Section: 3.1 · Task: drawing the as-is flow diagram

My contribution: I identified what the current flow Medtalk engineers might be using.

What it demonstrates: The diagram show the testing flow of the current MedTalk engineers are currently using.

<https://app.diagrams.net/>

W6-5 · Weekly reflection

Logged late · 5 min · 4/4 required parts · 31 Aug, 11:52

What did I personally complete this week?

I focused on technology and market research to understand similar sandboxes available. I reviewed comparable platforms, identified common features, and compared how they handle integration, compliance, and testing workflows. I also documented key requirements and gaps to help shape what our sandbox should include and how it should differ from existing options. Overall, I used the research to clarify the scope of the project and next steps.

What did I learn about the problem or the technology?

I learned that similar sandboxes often differ most in their supported integration methods, data validation approach, and operational constraints. Market research showed that success depends on clear onboarding, realistic test scenarios, and strong governance (security, privacy, and auditability). I also learned that defining success metrics early—such as latency, error handling, and reproducibility—helps guide technical decisions and prevents scope creep.

What decision did I influence, and why?

My research influenced the direction of the sandbox design by helping decide which features to prioritize first. I focused on capabilities that appear repeatedly across successful platforms (clear workflows, reliable API testing, and validation/auditing). This mattered because it ensures our build targets real user needs and avoids investing effort in less critical features. The comparisons also helped me justify trade-offs based on feasibility.

What will I finish before the next class?

Before next class, I will finalize the key requirements and translate the research into a short feature list and roadmap. I'll also draft a clearer problem statement and outline the proposed architecture or workflow for the sandbox. Finally, I will prepare any needed visuals or documentation (e.g., updated diagram) so we can review and confirm the plan in class.

Week 7

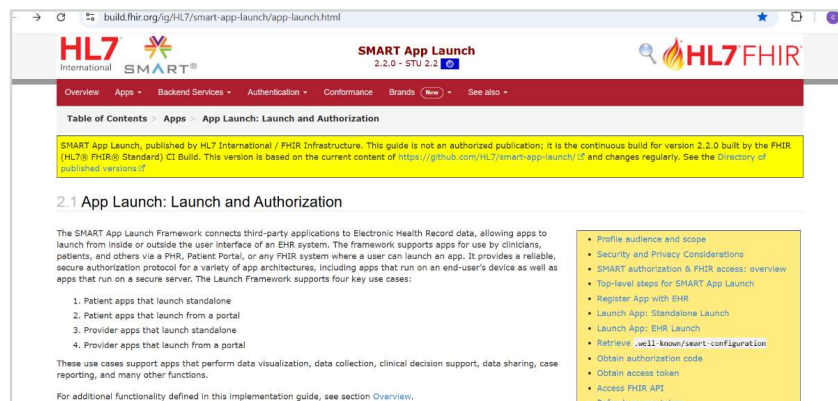
6 of 6 entries logged · 16 minutes recorded · 4 evidence items · 100% complete

W7-1 · Compare three existing tools

Logged late · 1 min · 4/4 required parts · 07 Sept, 10:34

TOOL / SYSTEM / APPROACH	VENDOR & LICENCE	FEATURES RELEVANT TO OUR P	LIMITATION WE CARE ABOUT	SOURCE
Epic sandbox	Free for developers to register and test apps against Epic's sandbox, but require a company	sandbox	Requires access under a company	https://fhir.epic.com/Developer/Index
SMART HEALTH IT SANDBOX	Generally an open developer/testing resource,	sandbox	non identified	https://build.fhir.org/g/HL7/smart-app-launch/app-launch.html
MediRecords	Commercial/proprietary software and API.	sandbox	requires signing up	https://support.mediarecords.com/hc/en-us/articles/13936381648399-MediRecords-Connect-Getting-Started-with-Connect-FHIR-APIs#choosing-connect-fhir

W7-1-01 · 07 SEPT 2026



Section: 3.1 · Task: Compare three existing tools

My contribution: Researching similar existing tools

What it demonstrates: The screenshot of the page that I got information on how implement smart sandbox

W7-2 · State the gap and your contribution

Logged late · 2 min · 4/4 required parts · 07 Sept, 11:02

The gap in existing solutions

The first gap is unsafe or limited integration testing versus a fully synthetic test environment. In the current state, MedTalk would rely on live partners, restricted sandboxes, or manual testing to validate integrations. That creates privacy and operational risk because real patient data cannot be used freely. The proposed solution closes this gap by creating a synthetic copy of the systems MedTalk connects to, so all testing happens without exposure to real patient records.

Your proposed technical contribution to closing it

The proposed technical solution is a synthetic healthcare integration and terminology test harness for MedTalk. It closes the main gaps by giving engineers a safe, reusable environment to simulate the systems MedTalk connects to, validate FHIR-based data exchange, and test terminology mapping without using any real patient data. This solution is well aligned with the Australian interoperability landscape. AU Core defines the minimum expectations for FHIR conformance and is intended to support testable system behavior. Australian Digital Health Agency guidance also identifies AU Core and AU CDI as the national baseline for FHIR connections to My Health Record, which reinforces the need for standards-based validation. Halo Connect is relevant as a production reference because it provides a FHIR facade for Best Practice Premier using AU Base and FHIR R4, but it is not enough on its own to support full synthetic testing and failure simulation

Technology or framework you selected

HL7 FHIR R4 AU Core AU Base AU CDI SNOMED CT-AU ICD-10-AM mapping layer MBS billing code mapping layer Halo Connect-compatible FHIR facade Postman / Newman for API test automation pytest or JUnit for automated test execution Docker for isolated synthetic environments WireMock or a similar mock-service framework for simulating delays, bad responses, and dropped connections OpenAPI / FHIR validation tools for schema checking JSON Schema validation for payload structure checks GitHub Actions / CI pipeline

One alternative you rejected, and why

Docker as the team do not have the skill on Docker skills: building and running containers, writing Dockerfiles, using Docker Compose, managing images and volumes

WireMock skills: creating mock endpoints, defining request/response stubs, simulating delays, errors, malformed payloads, and connection failures

CI/CD skills: setting up automated test runs in pipelines, usually with tools like GitHub Actions or Jenkins, and understanding test execution in build environments

W7-3 · Revise your figure — version 1 to version 2

Logged late · 1 min · 4/4 required parts · 07 Sept, 21:30

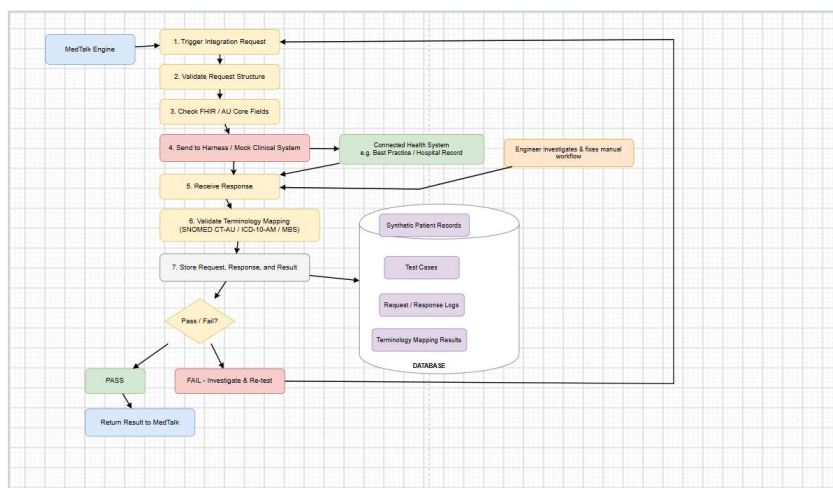
What you changed and why

The revised diagram makes the test harness an explicit intermediary between the MedTalk engine and the connected clinical system, rather than leaving the integration path implicit. It also adds a database layer to show where synthetic patient records, test cases, request/response logs, and terminology mapping results are stored for traceability and repeat testing. The flow is now broken into clearer stages, including request validation, FHIR/AU Core checking, terminology validation, response capture, and result storage, which makes the end-to-end process easier to explain in your report. Finally, the pass/fail decision is shown more clearly, with failures looping back for investigation and re-test, which better reflects an iterative integration testing workflow.

Whose feedback changed it

Sonam

W7-3-01 · 07 SEPT 2026

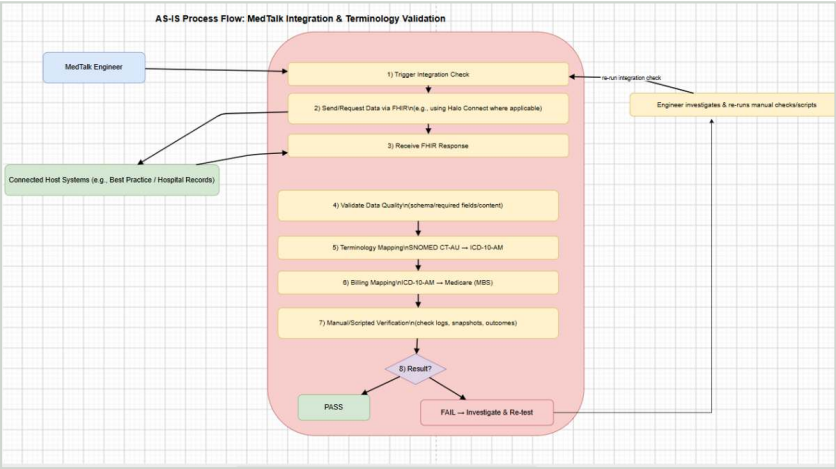


Section: 3.1 · Task: Revise your figure — version 1 to version 2

My contribution: I had also had to include the database to represent the best practice

What it demonstrates: The revised diagram of AS-IS flow

W7-3-02 · 07 SEPT 2026



Section: 3.1 · Task: Version 1

My contribution: I developed the AS-IS diagram based on the MedTalk documentation

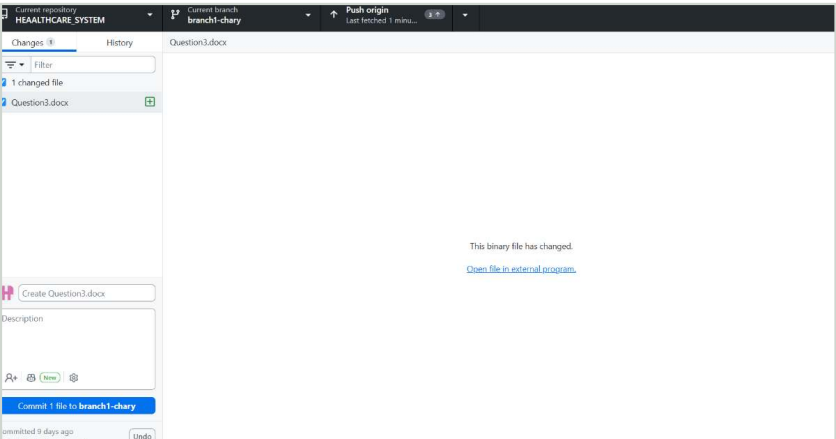
What it demonstrates: The initial diagram of the AS-IS diagram

W7-4 · Log a real GitHub action

Logged late · 9 min · 4/4 required parts · 07 Sept, 22:10

DATE	ISSUE / BRANCH / COMM	REFERENCE	WHAT YOU DID	LINK	OUTCOME
07SEP	branch1-chary	3	Writing the current situation of the FAZ Australia	https://github.com/SkinnySeal17/HEAALTHCARE_SYSTEM/tree/branch1-chary/	Part of the report done

W7-4-01 · 07 SEPT 2026



Section: github · Task: Log a real GitHub action

My contribution: Adding question 3 to my branch

What it demonstrates: This contains the literature review question

W7-5 · Record one technical decision

Logged late · 2 min · 4/4 required parts · 07 Sept, 23:18

Decision

To use python faker module to create fake synthetic patient for the proof of concept

Context — what forced the decision

Proof of concept, using the Python Faker module is a reasonable approach to generate synthetic patient data instead of building fully Australian-demographic data straight away. Faker is commonly used to create realistic but fictitious names, dates, addresses, and identifiers for testing and database bootstrapping, and it is suitable when the goal is to validate the harness workflow rather than population representativeness

Options considered

Synthea Synthea's main disadvantage is that it often produces data that is too clean and too uniform compared with real clinical records, so it may not reflect messy real-world variation, missing information, unusual treatment paths, or complex comorbidities. It can also be limited in how well it represents local healthcare patterns and code systems, which makes it less suitable when the goal is to mirror a specific national context such as Australian clinical workflows

Consequence, including what gets harder

The main consequence is that testing becomes less realistic, so it is harder to uncover messy real-world issues such as missing data, unusual care paths, comorbidities, and inconsistent coding. It also makes it harder to validate whether an integration or clinical documentation workflow will work reliably in actual healthcare settings.

W7-6 · Weekly reflection

Logged late · 1 min · 4/4 required parts · 08 Sept, 20:32

What did I personally complete this week?

This week, I completed the current-state and planning work for the MedTalk test harness project: refining the AS-IS workflow, identifying the integration gaps, and drafting the literature review structure for sections 3.1 to 3.4. I also worked on the synthetic patient data decision, comparing Faker and Synthea, and moved toward using Faker for the proof of concept because it better supports workflow validation and database testing.

What did I learn about the problem or the technology?

You learned that the problem is not just building a patient documentation workflow, but making it interoperable, standards-compliant, and realistic enough to test against real healthcare requirements. You also learned that AU Core, AU Base, and AU CDI form the Australian FHIR baseline for consistent data exchange, while SNOMED CT-AU and related terminology standards are needed to keep clinical coding accurate and usable in practice.

On the technology side, I learned that a test harness can help validate FHIR resources, terminology mapping, and workflow behavior before production, but that synthetic data tools like Synthea can be too uniform and may not capture messy real-world variation such as missing data, uncommon pathways, or inconsistent coding.

What decision did I influence, and why?

The decision to use Python Faker for the proof of concept instead of Synthea. The reason was that Faker is better suited to generating lightweight synthetic patient records for testing the integration workflow, database connectivity, and basic FHIR handling, while Synthea was seen as less suitable because it can be too heavy and too structured for this stage of the project.

What will I finish before the next class?

Before the next class, I will finish the AS-IS assessment, the literature review sections and the synthetic data/tool selection rationale for the MedTalk test harness project. Also I will complete the revised flow diagram and the initial proof-of-concept setup using Python Faker so the integration testing work is ready to present.

Week 8

6 of 7 entries logged · 13 minutes recorded · 5 evidence items · 89% complete

W8-1 · Show your component running

Logged late · 9 min · 4/4 required parts · 08 Sept, 21:06

What you built or configured

Synthetic data using Python Faker is a lightweight Python library used to generate realistic-looking synthetic data such as names, addresses, dates, and IDs for testing and prototyping. It is useful for creating safe placeholder patient records without using real information, which makes it suitable for integration testing and proof-of-concept work. This would then be used to as the data for the proof of concept test harness.

How to run or reproduce it

Python Faker can be used to generate synthetic test data by creating fake but realistic-looking values such as names, addresses, dates of birth, and email addresses. It is useful for proof-of-concept work because it allows safe testing of workflows and database connections without using real patient information.

W8-1-01 · 08 SEPT 2026

```

generate_au_data.py
109 def main():
110     random.seed(args.seed)
111     fake = Faker("en-AU")
112     fmt = args.format or ("json" if args.output.lower().endswith(
113         ".json") else "csv")
114     records = [generate_record(fake) for _ in range(args.num_reco
115
PS C:\Studies\Masters\S2\ICT942 Cyber security Project\Project docs\Python faker>
At line:1 char:1
+ /c:/Studies/Masters/S2/ICT942 Cyber security Project/Project docs\Pyt ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (c:/Studies/Masters/S2/ICT942:Str
ing) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Studies\Masters\S2\ICT942 Cyber security Project\Project docs\Python faker>
python generate_au_data.py -n 10 -o data.csv
Generated 10 synthetic Australian records -> data.csv (CSV)
PS C:\Studies\Masters\S2\ICT942 Cyber security Project\Project docs\Python faker>

```

Section: 4 · Task: Show your component running

My contribution: I developed synthetic data using python faker

What it demonstrates: The diagram is showing output after creating 10 patients

W8-1B · Upload your sitemap, wireframe and high-fidelity design

Logged late · 1 min · 7/7 required parts · 08 Sept, 21:57

Tool used

Figma

Sitemap — the screens or pages in your solution and how they connect

The solution is a test harness with three main parts: Home/Dashboard, FHIR Validation, and Fault Injection & Reporting. On the dashboard, users select the test type and choose synthetic patient datasets. The app then sends data through the FHIR Validation workflow, producing validation outcomes. Next, users can run Fault Injection scenarios to test resilience, with all results compiled into a Results/Report view, with navigation back to rerun tests.

Wireframe — which screen you wireframed and what a user can do on it

I wireframed the Dashboard/Test Harness screen. A user selects a synthetic patient dataset and test options, including whether to validate FHIR resources, map medical records to SNOMED, and enable fault injection. They can click Run to start the workflow. The screen includes a status/output panel showing validation results, mapping checks, and any injected fault outcomes. A reset option allows rerunning with new settings.

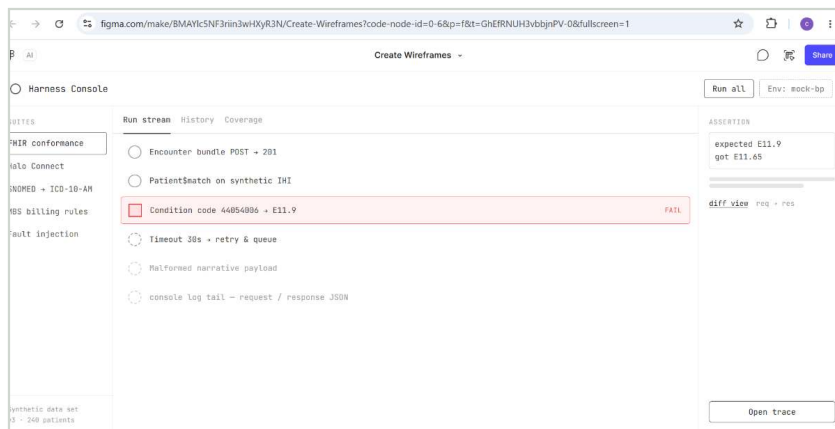
High-fidelity — what changed from the wireframe, and why

We have not yet created any high-fidelity mockups, so there are no final UI changes from the wireframe. What we currently have is a functional low-fidelity wireframe outlining the main inputs, actions (Run/Reset), and result areas. High-fidelity will be produced later after confirming branding, exact component sizes, and data visual layout requirements for validation and fault reports.

Which parts of these three artefacts did you personally produce?

I personally produced the **Sitemap** by defining the list of screens (Dashboard/Test Harness, FHIR Validation, Fault Injection & Reporting, and Results/Report) and how the navigation flows between them. I also personally produced the **wireframe** for the **Dashboard/Test Harness screen**, specifying what the user can do there—select synthetic patient data, choose validation and SNOMED mapping options, enable fault injection, and run/reset the tests—along with where test status and output results appear.

W8-1B-01 · 08 SEPT 2026

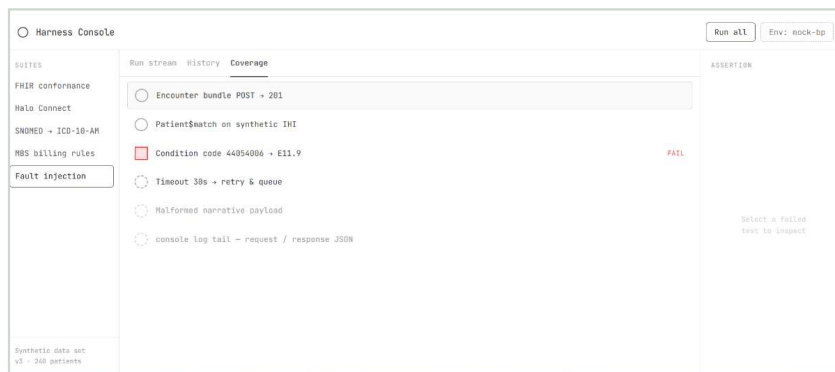


Section: 5 · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: I attempt to create the wireframe for the first page

What it demonstrates: is a wireframe for the test harness

W8-1B-02 · 08 SEPT 2026

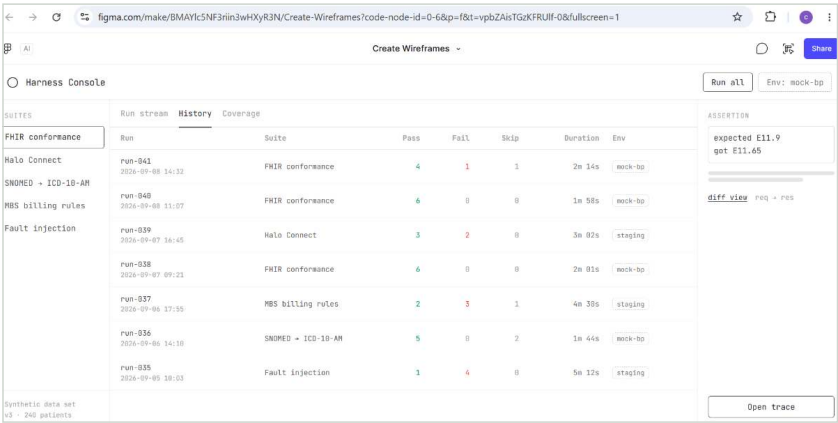


Section: 5 · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: I created the wireframe

What it demonstrates: This wireframe show when encounter selected then its a report to investigate failures

W8-1B-03 · 08 SEPT 2026



Section: 5 · Task: Upload your sitemap, wireframe and high-fidelity design

My contribution: I developed the wireframes

What it demonstrates: It shows the history page

W8-2 · Test it and record the results

Not started · 2 min · 1/4 required parts

No work recorded for this task.

W8-3 · Map your work to a security framework

Logged late · 1 min · 3/3 required parts · 08 Sept, 22:04

Framework used

NIST Cybersecurity Framework 2.0

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
AC-3	Access Enforcement	he test harness is designed to restrict who can run tests, view results, and manage configurations (e.g., only authorized users can start validation, mappings to SNOMED, or fault-injection scenarios). Role-based permissions can prevent unauthorized access to synthetic patient data and test outputs.	partial
AU-2	Event Logging	The harness should log security-relevant and test-relevant actions such as user logins, test runs, dataset selections, fault injection triggers, validation outcomes, and administrative changes. Logs enable traceability and investigation after failures or suspected misuse.	partial
SC-28	Protection of Information at Rest	The harness should protect stored synthetic patient data, generated reports, and validation artifacts using encryption at rest (and secure key management). This reduces the risk of data disclosure if storage is accessed or exfiltrated.	planned

W8-4 · Ethics, privacy, legal and AI governance

Logged late · 0 min · 3/3 required parts · 08 Sept, 22:05

Personal data your solution touches, and the obligation that applies

Our test harness uses synthetic patient data to validate FHIR resources and to map medical record terms against SNOMED. Although the data is synthetic, some datasets may still contain quasi-identifiers (e.g., dates, demographics, identifiers generated from real patterns), so we treat it as potentially personal information. Under the Privacy Act 1988 (Cth), we apply the Australian Privacy Principles (APPs)—specifically APP 11 (Security of personal information) for secure handling of test inputs, outputs, and logs, and APP 6 (Use/Disclosure) to ensure results are used only for validation and

testing purposes.

AI use in this project and how it was governed

This project uses no AI to make clinical judgments or to generate medical content. FHIR validation and SNOMED checks are rule/validator-based (e.g., conformance checks, schema validation, terminology matching), and fault injection is scripted to test robustness. For governance, we only allow AI tools (if any were used for development) under a documented workflow: outputs are reviewed and tested by us, we do not rely on AI-generated logic without unit tests, and we ensure no proprietary or sensitive real datasets are provided to external AI services.

One ethical or legal risk you cannot fully mitigate

A key residual risk is re-identification: even if data is labeled “synthetic,” it may be generated from templates that resemble real cohorts, or may include patterns that could allow linkage when combined with other information. This means we cannot fully guarantee the synthetic dataset will never be considered personal information under the Privacy Act. We mitigate this with access controls, encryption, minimised retention, and auditing, but legally/ethically we may still face uncertainty around identifiability in edge cases and future linkage attempts.

W8-5 · Own a slide

Logged late · 0 min · 5/5 required parts · 08 Sept, 22:21

Slide title and number you are responsible for

Industry and Technology Landscape Analysis

The three points you will make

In the Current State (As-Is) and the later technology/market, comparative, and gap sections, the assignment is mainly pushing you to highlight four things in sequence: first, what interoperability testing looks like today and where it falls short; second, what tools and technologies exist in the market now; third, how those solutions compare to your planned harness (what they don't cover or don't do consistently); and finally, the specific gaps that your solution will fill to move “As-Is” toward “To-Be” with repeatable, end-to-end validation, SNOMED checks, fault injection, and better reporting/evidence.

A question you expect from the audience, and your answer

Question: “How do you ensure synthetic patient data doesn't become misleading or inadvertently encode real-world biases?”

Answer: We ensure synthetic data is generated from controlled templates with defined distributions and constraints, and we document the generation method and assumptions. We evaluate outputs against validation and mapping rules to confirm coverage across edge cases (e.g., missing fields, rare codes). We also avoid claims about clinical correctness from synthesis alone and treat the harness as a technical interoperability and robustness tool. Where relevant, we use anonymisation-safe patterns and minimise retention to reduce re-identification or bias risks.

W8-5-01 · 08 SEPT 2026



Section: 3 · Task: Own a slide

My contribution: Industry and Technology Landscape Analysis

What it demonstrates: The powerpoint slide

W8-6 · Limitations, peer cross-check and final reflection

Logged late · 1 min · 7/7 required parts · 08 Sept, 22:27

Honest limitations of your contribution

A limitation of my contribution is that some parts of the work are still at a draft level (e.g., the early wireframe/screenshot and initial documentation). I also focused more on structuring the test harness workflow at a high level rather than producing fully detailed evidence for every requirement (e.g., complete governance mapping and comprehensive test scenario coverage). In addition, because we used synthetic patient data, some assumptions about edge cases may need further confirmation with additional test runs and refined datasets. Finally, parts of my analysis may need stronger links to specific artefact screenshots/logs once the final implementation is ready.

Which group member's evidence did you review today?

Rikesh diagrams

One specific thing that was strong and one thing that needs work in their evidence

A strong aspect was [specific strength—e.g., their clear gap-to-To-Be link and use of project-specific examples rather than generic statements]. Another area that needs work is [specific improvement—e.g., tightening word count/answers to exactly meet the rubric prompts, and adding more explicit how it applies to our harness detail and evidence screenshots/links]. Overall, the content direction was good, but it would score higher with more concrete references to what was implemented and the evidence supporting each claim.

What did I personally complete this week?

This week, I completed draft documentation and write-ups for the assignment areas focused on our solution's test harness workflow—including the As-Is/Market/Comparative/Gap narrative at a project-specific level. I also produced table-style summaries of key points (pain points, market tools, comparison gaps, and To-Be outcomes). Additionally, I helped draft security/ethics positioning by aligning it with the project activities (FHIR validation, SNOMED checks, and fault injection) and identifying the residual risks that can't be fully eliminated.

What did I learn about the problem or the technology?

I learned that interoperability testing is not just about FHIR conformance; it also depends heavily on terminology consistency (SNOMED mapping) and on resilience testing (how systems behave with malformed or faulty inputs). I also learned that many existing tools cover only one layer—either validation or general API testing—while our value comes from combining these layers into a single repeatable harness with structured reporting. Finally, I learned that “synthetic data” still requires careful governance because it may resemble real-world patterns and must be handled securely.

What decision did I influence, and why?

I influenced the decision to position our To-Be system as an integrated, repeatable harness rather than a collection of separate tools. The reason was that the As-Is review and market comparison showed fragmentation and inconsistent coverage in existing approaches. I also supported the decision to emphasize fault injection as a core feature, because it directly demonstrates robustness beyond schema correctness. This helps our project meet the assignment expectations for clear gaps, traceability, and governance-ready evidence.

What will I finish before the next class?

Before the next class, I will finalize my remaining write-ups to ensure each section is within the required word limits and clearly linked to our project (FHIR validation, SNOMED checks, fault injection, and reporting). I will also convert the draft points into clean slide/table-ready content and confirm that all claims are supported by the evidence we already have or will capture next. Finally, I will tighten any sections that currently read too broadly and add more explicit “what we did” details.

Literature and evidence tracker

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
6	Anantharaman, P., Shapiro, R. B., Varadharaju, V., & Locasto, M. E. (2023, November). A study of interoperability in electronic health record software. In Proceedings of the 2024 Workshop on Cybersecurity in Healthcare (pp. 53-60).	ACM Digital Library, IEEE Xplore,	interoperability protocols	Technology and market survey
6	C. L. Kim, 2017	ACM Digital Library, IEEE Xplore,	Healthcare systems	Current situation as-is
6	Diouf, I., Grimes, J., O'Brien, M. J., Hassanzadeh, H., Truran, D., Ngo, H., ... & Reguant, R. (2024,)	ACM Digital Library, IEEE Xplore,	Australian Synthetic data creation	For the project to create synthetic data
7	Epic sandbox (ree for developers to register and test apps against Epic's sandbox. but require a company)	https://fhir.epic.com/Developer/Index	sandbox	Requires access under a company
7	SMART HEALTH IT SANDBOX (Generally an open developer/testing resource,)	https://build.fhir.org/ig/HL7/smart-app-launch/app-launch.html	sandbox	non identified
7	MediRecords (Commercial/proprietary software and API.)	https://support.medirecords.com/hc/en-us/articles/13936381648399-MediRecords-Connect-Getting-Started-with-Connect-FHIR-APIs#choosing-connect-fhir	sandbox	requires signing up
8	NIST Cybersecurity Framework 2.0 AC-3	framework document	Access Enforcement	he test harness is designed to restrict who can run tests, view results, and manage configurations (e.g., only authorized users can start validation, mappings to SNOMED, or fault-injection scenarios). Role-based permissions can prevent unauthorized access to synthetic patient data and test outputs.
8	NIST Cybersecurity Framework 2.0 AU-2	framework document	Event Logging	The harness should log security-relevant and test-relevant actions such as user logins, test runs, dataset selections, fault injection triggers, validation outcomes, and administrative changes. Logs enable traceability and investigation after failures or suspected misuse.

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
8	NIST Cybersecurity Framework 2.0 SC-28	framework document	Protection of Information at Rest	The harness should protect stored synthetic patient data, generated reports, and validation artifacts using encryption at rest (and secure key management). This reduces the risk of data disclosure if storage is accessed or exfiltrated.

GitHub and project-management activity

DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
07SEP	GitHub	branch1-chary 3	Writing the current situation of the FAZ Australia	https://github.com/SkinnySeal17/HEAALTHCARE_SYSTEM/tree/branch1-chary/	Part of the report done
Wk6	Board	3.1 Current State Assessment: As-Is Anal	Created or took ownership		
Wk6	Board	3.2 Technology and Market Survey	Created or took ownership		
Wk6	Board	3.3 Comparative Analysis of Existing Sol	Created or took ownership		
Wk6	Board	3.4 Gap Analysis: As-Is to To-Be	Created or took ownership		

Evidence index

NO.	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W6-2-02	31 Aug 2026	3.3	Literature search log	This a CSIRO journal on how to create Australian patient synthetic data
W6-2-03	31 Aug 2026	3.1	Literature search log	This will be part of understanding the healthcare environment and workflows.
W6-2-07	31 Aug 2026	3.1	Literature search log	Explores the cyberattacks popular in healthcare system
W6-3-01	31 Aug 2026	3	Claim your tasks on the board	The jira project assignment distribution of tasks
W6-4-01	31 Aug 2026	3.1	Draw the first figure	The diagram show the testing flow of the current MedTalk engineers are currently using.
W7-1-01	07 Sept 2026	3.1	Compare three existing tools	The screenshot of the page that I got information on how implement smart sandbox
W7-3-01	07 Sept 2026	3.1	Revise your figure — version 1 to version 2	The revised diagram of AS-Is flow
W7-3-02	07 Sept 2026	3.1	Revise your figure — version 1 to version 2	The initial diagram of the AS-IS diagram
W7-4-01	07 Sept 2026	gitHub	Log a real GitHub action	This contains the literature review question
W8-1-01	08 Sept 2026	4	Show your component running	The diagram is showing output after creating 10 patients
W8-1B-01	08 Sept 2026	5	Upload your sitemap, wireframe and high-fidelity design	is a wireframe for the test harness

NO .	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W8-1B-02	08 Sept 2026	5	Upload your sitemap, wireframe and high-fidelity design	This wireframe show when encounter selected then its a report to investigate failures
W8-1B-03	08 Sept 2026	5	Upload your sitemap, wireframe and high-fidelity design	It shows the history page
W8-5-01	08 Sept 2026	3	Own a slide	The powerpoint slide

VERIFICATION

Time log

Each entry below was recorded automatically while the task was open in the portfolio tool. The timer pauses after three minutes without activity.

TASK	SESSION START	DURATION	CONTEXT
S-1	24 Aug, 10:17	00:15	In class
S-2	24 Aug, 10:28	00:30	In class
S-2	24 Aug, 10:28	00:17	In class
S-1	24 Aug, 10:32	00:30	In class
S-1	24 Aug, 10:32	00:28	In class
S-2	24 Aug, 10:33	00:28	In class
W6-1	24 Aug, 10:35	00:30	Outside class
W6-1	24 Aug, 10:36	00:30	Outside class
W6-1	24 Aug, 10:36	00:31	Outside class
W6-1	24 Aug, 10:37	00:31	Outside class
W6-1	24 Aug, 10:38	00:31	Outside class
W6-1	24 Aug, 10:38	00:31	Outside class
W6-1	24 Aug, 10:39	00:26	Outside class
W6-2	24 Aug, 10:55	00:27	Outside class
W6-3	24 Aug, 11:08	00:30	Outside class
W6-3	24 Aug, 11:09	00:31	Outside class
W6-3	24 Aug, 11:09	00:30	Outside class
W6-4	24 Aug, 11:10	00:20	Outside class
W6-5	24 Aug, 11:10	00:30	Outside class
W6-5	24 Aug, 11:10	00:30	Outside class
W6-5	24 Aug, 11:11	00:31	Outside class
W6-5	24 Aug, 11:11	00:06	Outside class
S-2	31 Aug, 08:58	00:30	Outside class
S-2	31 Aug, 08:58	00:30	Outside class
S-2	31 Aug, 08:59	00:31	Outside class
S-2	31 Aug, 08:59	00:17	Outside class
W6-1	31 Aug, 09:07	00:30	In class
W6-1	31 Aug, 09:07	00:30	In class
W6-1	31 Aug, 09:08	00:31	In class
W6-1	31 Aug, 09:08	00:30	In class
W6-1	31 Aug, 09:09	00:31	In class
W6-1	31 Aug, 09:09	00:30	In class
W6-1	31 Aug, 09:10	00:31	In class

TASK	SESSION START	DURATION	CONTEXT
W6-1	31 Aug, 09:10	00:31	In class
W6-1	31 Aug, 09:11	00:31	In class
W6-1	31 Aug, 09:11	00:30	In class
W6-1	31 Aug, 09:12	00:30	In class
W6-1	31 Aug, 09:12	00:31	In class
W6-1	31 Aug, 09:13	00:30	In class
W6-1	31 Aug, 09:13	00:30	In class
W6-1	31 Aug, 09:14	00:10	In class
W6-2	31 Aug, 09:14	00:30	In class
W6-2	31 Aug, 09:15	00:12	In class
W6-3	31 Aug, 09:43	00:30	In class
W6-3	31 Aug, 09:44	00:30	In class
W6-3	31 Aug, 09:44	00:30	In class
W6-3	31 Aug, 09:45	00:30	In class
W6-3	31 Aug, 09:45	00:30	In class
W6-3	31 Aug, 09:46	00:31	In class
W6-3	31 Aug, 09:46	00:23	In class
W6-4	31 Aug, 10:15	00:30	In class
W6-4	31 Aug, 10:16	00:30	In class
W6-4	31 Aug, 10:16	00:31	In class
W6-4	31 Aug, 10:17	00:30	In class
W6-4	31 Aug, 10:17	00:31	In class
W6-4	31 Aug, 10:18	00:31	In class
W6-4	31 Aug, 10:18	00:16	In class
W6-4	31 Aug, 10:34	00:05	In class
W6-4	31 Aug, 10:34	00:05	In class
W6-4	31 Aug, 10:34	00:30	In class
W6-4	31 Aug, 10:35	00:31	In class
W6-4	31 Aug, 10:35	00:30	In class
W6-4	31 Aug, 10:36	00:31	In class
W6-4	31 Aug, 10:36	00:31	In class
W6-4	31 Aug, 10:37	00:31	In class
W6-1	31 Aug, 10:44	00:30	In class
W6-1	31 Aug, 10:44	00:31	In class
W6-1	31 Aug, 10:45	00:30	In class
W6-1	31 Aug, 10:45	00:30	In class
W6-1	31 Aug, 10:46	00:31	In class

TASK	SESSION START	DURATION	CONTEXT
W6-1	31 Aug, 10:46	00:31	In class
W6-1	31 Aug, 10:47	00:31	In class
W6-1	31 Aug, 10:47	00:31	In class
W6-1	31 Aug, 10:48	00:31	In class
W6-1	31 Aug, 10:48	00:31	In class
W6-1	31 Aug, 10:49	00:31	In class
W6-1	31 Aug, 10:49	00:31	In class
W6-1	31 Aug, 10:50	00:31	In class
W6-1	31 Aug, 10:50	00:31	In class
W6-1	31 Aug, 10:51	00:31	In class
W6-1	31 Aug, 10:51	00:31	In class
W6-1	31 Aug, 10:52	00:31	In class
W6-1	31 Aug, 10:52	00:30	In class
W6-1	31 Aug, 10:53	00:06	In class
W6-2	31 Aug, 10:58	00:25	In class
W6-3	31 Aug, 10:59	00:21	In class
W6-4	31 Aug, 10:59	00:30	In class
W6-4	31 Aug, 10:59	00:31	In class
W6-4	31 Aug, 11:00	00:30	In class
W6-4	31 Aug, 11:01	00:31	In class
W6-4	31 Aug, 11:01	00:31	In class
W6-4	31 Aug, 11:02	00:31	In class
W6-4	31 Aug, 11:02	00:21	In class
W6-5	31 Aug, 11:48	00:30	Outside class
W6-5	31 Aug, 11:48	00:30	Outside class
W6-5	31 Aug, 11:49	00:31	Outside class
W6-5	31 Aug, 11:49	00:31	Outside class
W6-5	31 Aug, 11:50	00:30	Outside class
W6-5	31 Aug, 11:50	00:31	Outside class
W6-5	31 Aug, 11:51	00:30	Outside class
W6-5	31 Aug, 11:51	00:18	Outside class
W7-1	07 Sept, 09:45	00:11	Outside class
W7-2	07 Sept, 10:34	00:30	Outside class
W7-2	07 Sept, 10:34	00:31	Outside class
W7-2	07 Sept, 10:35	00:31	Outside class
W7-2	07 Sept, 10:35	00:31	Outside class
W7-1	07 Sept, 11:02	00:27	Outside class

TASK	SESSION START	DURATION	CONTEXT
W7-2	07 Sept, 11:03	00:08	Outside class
W7-3	07 Sept, 11:03	00:30	Outside class
W7-3	07 Sept, 11:04	00:30	Outside class
W7-3	07 Sept, 11:04	00:08	Outside class
W7-4	07 Sept, 21:31	00:30	Outside class
W7-4	07 Sept, 21:32	00:31	Outside class
W7-4	07 Sept, 21:32	00:30	Outside class
W7-4	07 Sept, 21:33	00:30	Outside class
W7-4	07 Sept, 21:33	00:31	Outside class
W7-4	07 Sept, 21:34	00:31	Outside class
W7-4	07 Sept, 21:34	00:30	Outside class
W7-4	07 Sept, 21:35	00:24	Outside class
W7-5	07 Sept, 21:35	00:19	Outside class

Weekly individual reflection

Week 6

What did I personally complete this week?

I focused on technology and market research to understand similar sandboxes available. I reviewed comparable platforms, identified common features, and compared how they handle integration, compliance, and testing workflows. I also documented key requirements and gaps to help shape what our sandbox should include and how it should differ from existing options. Overall, I used the research to clarify the scope of the project and next steps.

What did I learn about the problem or the technology?

I learned that similar sandboxes often differ most in their supported integration methods, data validation approach, and operational constraints. Market research showed that success depends on clear onboarding, realistic test scenarios, and strong governance (security, privacy, and auditability). I also learned that defining success metrics early—such as latency, error handling, and reproducibility—helps guide technical decisions and prevents scope creep.

What decision did I influence, and why?

My research influenced the direction of the sandbox design by helping decide which features to prioritize first. I focused on capabilities that appear repeatedly across successful platforms (clear workflows, reliable API testing, and validation/auditing). This mattered because it ensures our build targets real user needs and avoids investing effort in less critical features. The comparisons also helped me justify trade-offs based on feasibility.

What will I finish before the next class?

Before next class, I will finalize the key requirements and translate the research into a short feature list and roadmap. I'll also draft a clearer problem statement and outline the proposed architecture or workflow for the sandbox. Finally, I will prepare any needed visuals or documentation (e.g., updated diagram) so we can review and confirm the plan in class.

Week 7

What did I personally complete this week?

This week, I completed the current-state and planning work for the MedTalk test harness project: refining the AS-IS workflow, identifying the integration gaps, and drafting the literature review structure for sections 3.1 to 3.4. I also worked on the synthetic patient data decision, comparing Faker and Synthea, and moved toward using Faker for the proof of concept because it better supports workflow validation and database testing.

What did I learn about the problem or the technology?

You learned that the problem is not just building a patient documentation workflow, but making it interoperable, standards-compliant, and realistic enough to test against real healthcare requirements. You also learned that AU Core, AU Base, and AU CDI form the Australian FHIR baseline for consistent data exchange, while SNOMED CT-AU and related terminology standards are needed to keep clinical coding accurate and usable in practice.

On the technology side, I learned that a test harness can help validate FHIR resources, terminology mapping, and workflow behavior before production, but that synthetic data tools like Synthea can be too uniform and may not capture messy real-world variation such as missing data, uncommon pathways, or inconsistent coding.

What decision did I influence, and why?

The decision to use Python Faker for the proof of concept instead of Synthea. The reason was that Faker is better suited to generating lightweight synthetic patient records for testing the integration workflow, database connectivity, and basic FHIR handling, while Synthea was seen as less suitable because it can be too heavy and too structured for this stage of the project.

What will I finish before the next class?

Before the next class, I will finish the AS-IS assessment, the literature review sections and the synthetic data/tool selection rationale for the MedTalk test harness project. Also I will complete the revised flow diagram and the initial proof-of-concept setup using Python Faker so the integration testing work is ready to present.

Week 8

What did I personally complete this week?

This week, I completed draft documentation and write-ups for the assignment areas focused on our solution's test harness

workflow—including the As-Is/Market/Comparative/Gap narrative at a project-specific level. I also produced table-style summaries of key points (pain points, market tools, comparison gaps, and To-Be outcomes). Additionally, I helped draft security/ethics positioning by aligning it with the project activities (FHIR validation, SNOMED checks, and fault injection) and identifying the residual risks that can't be fully eliminated.

What did I learn about the problem or the technology?

I learned that interoperability testing is not just about FHIR conformance; it also depends heavily on terminology consistency (SNOMED mapping) and on resilience testing (how systems behave with malformed or faulty inputs). I also learned that many existing tools cover only one layer—either validation or general API testing—while our value comes from combining these layers into a single repeatable harness with structured reporting. Finally, I learned that “synthetic data” still requires careful governance because it may resemble real-world patterns and must be handled securely.

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I influenced the decision to position our To-Be system as an integrated, repeatable harness rather than a collection of separate tools. The reason was that the As-Is review and market comparison showed fragmentation and inconsistent coverage in existing approaches. I also supported the decision to emphasize fault injection as a core feature, because it directly demonstrates robustness beyond schema correctness. This helps our project meet the assignment expectations for clear gaps, traceability, and governance-ready evidence.

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Declaration

I confirm that the work recorded in this portfolio is my own individual contribution to the group project. Screenshots and links refer to work I personally performed. Where AI tools were used, this is disclosed in the Week 8 governance entry.

Signed: _____ Date: _____